Lab 01 – Basic Subnetting and Routing

A beginner-friendly lab showing how subnetting controls traffic flow between computers.

What This Lab Was About

The goal of this lab was to understand how computers decide whether network traffic stays local or must be sent to a router. Two Ubuntu virtual machines were used to observe this behavior.

Lab Setup

Two Ubuntu virtual machines were created using VMware Workstation Pro. Both machines were connected to a Host-only virtual network, which acts like a virtual switch.

Network Layout

Ubuntu-PC1 used the IP address 192.168.1.10/24. Ubuntu-PC2 used 192.168.1.20/24 at first. Later, Ubuntu-PC2 was moved to 192.168.2.20/24 to place it in a different subnet.

What Went Wrong (Initial Failure)

At first, the computers could not communicate. One virtual machine was not connected to the same Host-only network, meaning the machines were not on the same virtual switch. Because of this, even though the IP addresses looked correct, traffic had nowhere to go.

Why Subnetting Broke the Connection

When Ubuntu-PC2 was moved to a different subnet, Ubuntu-PC1 determined the destination was not local. The system attempted to send the traffic to a router, but no default gateway had been configured, causing communication to fail.

How It Was Fixed

The network adapter was corrected so both machines shared the same virtual switch. Ubuntu-PC1 was then given a second IP address and configured to forward traffic, allowing it to act as a router. A default gateway was added to Ubuntu-PC2.

Final Result

After routing was configured, traffic successfully flowed between the two subnets. This confirmed that subnetting controls whether traffic stays local or must be routed.

What I Learned

Subnetting tells a computer whether another device is local or remote. Devices on the same subnet communicate directly, while devices on different subnets require a router and a default gateway.